Research question: How does the quality and quantity of training data impact the performance of a language model on low-resource languages?

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We met with our project supervisor two weeks ago to go over the scope of the study. As none of us have any ML background at all, he provided us with a list of the technical topics we needed to get our heads around first. We've now done that and as a result we're in a position to read the papers with a much better understanding.

1. Doing the article search. We searched Google Scholar and used Claude (the AI assistant) to suggest candidate papers, which we then checked against their references. We started with the first four references given and used terms such as *low-resource languages, data quality, data quantity, fine-tuning, continued pretraining* and *multilingual language models*. So far we've got four papers selected: Ogueji et al. (2021, *Small Data? No Problem!*), Kreutzer et al. (2022, *Quality at a Glance*), Chang et al. (2024, *When Is Multilinguality a Curse?*) and Szep et al. (2026, *Fine-tuning Large Language Models with Limited Data*).

2. Organising the articles and notes. We all have our own system. Jayesh has a GitHub repo with one Markdown review per paper, a reading list with a search log, a BibTeX file and a findings file which collects the conclusions from the various papers. Moksha has handwritten notes. We go through our notes when we meet to compare.

3. First reading and key concepts. We're currently in the process of doing the first reading of our four articles. We're going through each one (abstract, introduction, results and conclusion) to identify the key terms and to decide which of the articles we're going to keep. The key concepts we're using to read them are the technical topics that our supervisor has provided us with: language modelling, pretraining (masked and causal), continued pretraining, fine tuning and parameter efficient methods and evaluation metrics.

4. Current understanding, uncertainties and priorities. Our current understanding is confined to the technical context: how the models are trained, how they can be adapted to a new language and how their performance is assessed. We haven't actually come to any conclusions from the papers themselves. Our initial questions are: how do we compare results across papers that use different metrics? What does 'low-resource' actually mean? How do we measure data quality for these languages?

The next stage is to present two of the papers each to our supervisor on Thursday, then do the first reading of the rest and move on to the first four references.